# Extended

08/23/2026 18:06:44
https://gemini.google.com/app/60bac5f761e3b414

**Defining the Problem** I've successfully identified the core challenge:...

solve k in detailed

**Defining the Problem** I've successfully identified and formulated the...

Solve l in detailed

**Defining the Problem** I've precisely defined the inversion problem,...

Solve m in detailed

**Defining the Scope** I have clearly defined the objective and identifie...

Solve n in detailed

**Defining the Problem** I've successfully identified the specific problem...

Assignment 2.docx

Solve it in detailed

*23/08/2026, 10:40:18*

**Thinking steps**

**Defining the Scope**

I have successfully identified and listed all of the problems from the images that need solutions.

**Structuring Solutions**

I'm systematically tackling each problem, with structured, detailed solutions for all images provided, starting with 1.16.

**Analyzing System Properties**

I've determined linearity, time-invariance, and derived the bounded output for problems 1.18 and 1.19.

**Assessing Linearity**

I've determined linearity and time-invariance for the provided systems in problems 1.18, 1.19, and completed 1.20.

**Evaluating Invertibility**

I have successfully assessed the invertibility of several systems within problem 1.30.

**Determining Invertibility**

I have carefully assessed invertibility for several new systems, providing clear reasoning.

I have successfully assessed the invertibility of several systems within problem 1.30.

**Evaluating Invertibility**

I have carefully assessed invertibility for several new systems, providing clear reasoning, now working on markdown format.

**Improving the Formatting**

I've standardized variable notation and refined the markdown, especially with headers and citations.

**Assessing System Properties**

I've determined linearity, time-invariance, and boundedness for problems 1.18 and 1.19, with clear calculations.

**Assessing System Properties**

I've determined linearity and time-invariance, carefully explaining odd function and typos.

I have completed the linearity and time invariance assessment, successfully solving problem 1.30.

**Determining Invertibility**

I have carefully assessed system invertibility in (c)-(m), providing clear and concise reasoning.

**Finalizing Assessments**

I determined invertibility for the system described, and reviewed formatting and citation rules to ensure conciseness.

**Analyzing System Properties**

I have now determined linearity and time-invariance for the odd function system with detailed calculations and justifications.

✓ Done

Here are the detailed step-by-step solutions for each of the problems presented in the images.

## Problem 1.16

The input-output relationship for the discrete-time system is given as $y[n] = x[n]x[n-2]$[cite: 1].

- **(a) Is the system memoryless?**
  No[cite: 1]. A system is memoryless if its output at any time $n$ depends only on the input at that exact same time n$[cite: 1]. In this case, the output $y[n] depends on the present input $x[n]$ and a past input $x[n-2]$[cite: 1].
- **(b) Determine the output when the input is $x[n] = A\delta[n]$:**
  Substituting $x[n]$ into the system equation yields y[n] = (A\delta[n])(A\delta[n-2]) = A^2 \delta[n]\delta[n-2]$[cite: 1]. The unit impulse function $\delta[n] is non-zero only at $n = 0$, while $\delta[n-2]$ is non-zero only at n = 2$[cite: 1]. Because

there is no integer $n where both terms are simultaneously non-zero, their product is always zero[cite: 1]. Therefore, $y[n] = 0$ for all $n$[cite: 1].

- **(c) Is the system invertible?**
  No[cite: 1]. A system is invertible if distinct inputs produce distinct outputs, allowing the input to be uniquely recovered[cite: 1]. As shown in part (b), the input $x[n] = A\delta[n]$ produces the output $y[n] = 0$ regardless of the value of A$[cite: 1]. Because multiple different inputs (e.g., $A=1 and $A=2$) map to the exact same output, the system is not invertible[cite: 1].

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## Problem 1.17

The continuous-time system is defined as $y(t) = x(\sin(t))$[cite: 1].

- **(a) Is this system causal?**
  No[cite: 1]. A system is causal if the output at time $t$ depends only on inputs at times $\tau \le t$[cite: 1]. Let us evaluate the system at $t = -\pi$[cite: 1]. The output is $y(-\pi) = x(\sin(-\pi)) = x(0)$[cite: 1]. Since the output at a past time ($t = -\pi$) depends on an input at a future time ($t = 0$), the system is non-causal[cite: 1].
- **(b) Is this system linear?**
  Yes[cite: 1]. Let $x_1(t)$ produce $y_1(t) = x_1(\sin(t))$ and $x_2(t)$ produce y_2(t) = x_2(\sin(t))$[cite: 1]. If we apply a linear combination as the input, $x_3(t) = ax_1(t) + bx_2(t), the output is $y_3(t) = x_3(\sin(t)) = ax_1(\sin(t)) + bx_2(\sin(t)) = ay_1(t) + by_2(t)$[cite: 1]. This satisfies the principle of superposition[cite: 1].

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## Problem 1.18

The discrete-time system is defined as $y[n] = \sum_{k=n-n_0}^{n+n_0} x[k]$[cite: 1].

- **(a) Is this system linear?**
  Yes[cite: 1]. The summation operator is linear[cite: 1]. For an input combination $ax_1[n] + bx_2[n]$, the output is $\sum_{k=n-n_0}^{n+n_0} (ax_1[k] + bx_2[k]) = a\sum x_1[k] + b\sum x_2[k] = ay_1[n] + by_2[n]$[cite: 1].
- **(a) Is this system time-invariant?** (Note: Labeled 'a' in the image, corresponding to the second question).

Yes[cite: 1]. Let the input be shifted by $N$ such that x_1[n] = x[n-N]$[cite: 1]. The output is $y_1[n] = \sum_{k=n-n_0}^{n+n_0} x[k-N]$[cite: 1]. Let $m = k-N, meaning the summation bounds shift to $(n-N)-n_0$ and (n-N)+n_0$[cite: 1]. The output becomes $\sum_{m=(n-N)-n_0}^{(n-N)+n_0} x[m], which equals y[n-N]$[cite: 1]. Since $y_1[n] = y[n-N], the system is time-invariant[cite: 1].

- **(c) Express $C$ in terms of $B$ and $n_0$:**
  We are given that $|x[n]| < B$ for all n$[cite: 1]. We want to bound $|y[n]| = \left| \sum_{k=n-n_0}^{n+n_0} x[k] \right|$[cite: 1]. By the triangle inequality, $|y[n]| \le \sum_{k=n-n_0}^{n+n_0} |x[k]|$[cite: 1]. Substituting the bound $B, we get $|y[n]| < \sum_{k=n-n_0}^{n+n_0} B$[cite: 1]. The number of terms in this summation is $(n+n_0) - (n-n_0) + 1 = 2n_0 + 1$[cite: 1]. Therefore, $|y[n]| < (2n_0 + 1)B$[cite: 1]. We conclude that $C = (2n_0 + 1)B$[cite: 1].

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## Problem 1.19

- **(a)** $y(t) = t^2 x(t-1)$[cite: 1]
  - **Linear**: Yes, because $t^2 (ax_1(t-1) + bx_2(t-1)) = ay_1(t) + by_2(t)$[cite: 1].
  - **Time-invariant**: No[cite: 1]. A delayed input $x_1(t) = x(t-t_0)$ yields y_1(t) = t^2 x(t-1-t_0)$[cite: 1]. However, delaying the output gives $y(t-t_0) = (t-t_0)^2 x(t-t_0-1)$[cite: 1]. Since $y_1(t) \neq y(t-t_0), it is time-varying[cite: 1].
- **(b)** $y[n] = x^2[n-2]$[cite: 1]
  - **Linear**: No[cite: 1]. If the input is scaled by $c$, the output is $(cx[n-2])^2 = c^2x^2[n-2] = c^2y[n]$, which does not equal $cy[n]$ for arbitrary $c$[cite: 1].
  - **Time-invariant**: Yes[cite: 1]. Delaying the input by $N$ gives x_1[n] = x[n-N]$[cite: 1]. The output is $y_1[n] = x^2[n-N-2], which precisely matches $y[n-N]$[cite: 1].
- **(c)** $y[n] = x[n+1] - x[n-1]$[cite: 1]
  - **Linear**: Yes[cite: 1]. The operations of shifting and subtracting are linear[cite: 1].

    - **Time-invariant**: Yes[cite: 1]. Delaying the input by $N$ yields $x[n-N+1] - x[n-N-1]$, matching the delayed output $y[n-N]$[cite: 1].

- **(d)** $y[n] = \mathcal{O}d{x[n]} = \frac{x[n] - x[-n]}{2}$[cite: 1] *(Assuming standard "Odd part" operator and resolving the mixed variable typo in the image)*

    - **Linear**: Yes[cite: 1]. $\mathcal{O}d{a x_1 + b x_2} = a\mathcal{O}d{x_1} + b\mathcal{O}d{x_2}$[cite: 1].

    - **Time-invariant**: No[cite: 1]. Delaying the input by $N$ yields $y_1[n] = \frac{x[n-N] - x[-n-N]}{2}$[cite: 1]. Delaying the output gives $y[n-N] = \frac{x[n-N] - x[-(n-N)]}{2} = \frac{x[n-N] - x[-n+N]}{2}$[cite: 1]. These two equations are not equivalent[cite: 1].

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## Problem 1.20

We are given a continuous-time linear system $S$ where $S\{e^{j2t}\} = e^{j3t}$ and $S{e^{-j2t}} = e^{-j3t}$[cite: 1].

- **(a) Determine output for $x_1(t) = \cos(2t)$:**
Using Euler's formula, we can rewrite the input as $x_1(t) = \frac{1}{2} e^{j2t} + \frac{1}{2} e^{-j2t}$[cite: 1]. Because the system is linear, we can process each term separately[cite: 1]. $y_1(t) = \frac{1}{2} S{e^{j2t}} + \frac{1}{2} S{e^{-j2t}} = \frac{1}{2} e^{j3t} + \frac{1}{2} e^{-j3t}$[cite: 1]. Converting back using Euler's formula gives $y_1(t) = \cos(3t)$[cite: 1].

- **(b) Determine output for $x_2(t) = \cos(2(t-1/2))$:**
First, rewrite the input: \cos(2t - 1) = \frac{1}{2} e^{j(2t-1)} + \frac{1}{2} e^{-j(2t-1)} = \frac{e^{-j}}{2} e^{j2t} + \frac{e^{j}}{2} e^{-j2t}$[cite: 1]. Applying the linear system $S yields $y_2(t) = \frac{e^{-j}}{2} e^{j3t} + \frac{e^{j}}{2} e^{-j3t} = \frac{1}{2} e^{j(3t-1)} + \frac{1}{2} e^{-j(3t-1)}$[cite: 1]. This simplifies to $y_2(t) = \cos(3t-1)$[cite: 1].

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## Problem 1.30

- **(a)** $y(t) = x(t-4)$[cite: 1]. **Invertible**[cite: 1]. Inverse system is $x(t) = y(t+4)$[cite: 1].

- **(b)** y(t) = \cos[x(t)]$[cite: 1]. **Not invertible**[cite: 1]. Inputs $x_1(t) = 0 and $x_2(t) = 2\pi$ both result in $y(t) = 1$, so the input cannot be uniquely determined[cite: 1].

- **(c)** y[n] = nx[n]$[cite: 1]. **Not invertible**[cite: 1]. At $n=0, $y[0] = 0 \cdot x[0] = 0$ regardless of x[0]$[cite: 1]. Inputs $x_1[n] = \delta[n] and $x_2[n] = 2\delta[n]$ both yield $y[n] = 0$ everywhere[cite: 1].
- **(d)** $y(t) = \int_{-\infty}^{t} x(\tau) d\tau$[cite: 1]. **Invertible**[cite: 1]. Inverse system is the differentiator $x(t) = \frac{dy(t)}{dt}$[cite: 1].
- **(e)** $y[n] = x[n-1]$ for n \ge 1$; $y[0] = 0$; $y[n] = x[n] for n \le -1$[cite: 1]. **Invertible**[cite: 1]. Inverse system: $x[n] = y[n+1] for $n \geq 0$, and $x[n] = y[n]$ for $n \le -1$[cite: 1].
- **(f)** y[n] = x[n]x[n-1]$[cite: 1]. **Not invertible**[cite: 1]. The constant input sequences $x_1[n] = 1 and $x_2[n] = -1$ both yield the exact same output $y[n] = 1$[cite: 1].
- **(g)** $y[n] = x[1-n]$[cite: 1]. **Invertible**[cite: 1]. Inverse system is $x[n] = y[1-n]$[cite: 1].
- **(h)** $y(t) = \int_{-\infty}^{t} e^{-(t-\tau)} x(\tau) d\tau$[cite: 1]. **Invertible**[cite: 1]. Using Leibniz's rule to differentiate gives $\frac{dy(t)}{dt} = x(t) - y(t)$[cite: 1]. Rearranging terms, the inverse system is $x(t) = \frac{dy(t)}{dt} + y(t)$[cite: 1].
- **(i)** $y[n] = \sum_{k=-\infty}^{n} (1/2)^{n-k} x[k]$[cite: 1]. **Invertible**[cite: 1]. Expanding the sum reveals a recursive relationship: $y[n] = x[n] + \frac{1}{2}y[n-1]$[cite: 1]. The inverse system is $x[n] = y[n] - \frac{1}{2}y[n-1]$[cite: 1].
- **(j)** y(t) = \frac{dx(t)}{dt}$[cite: 1]. **Not invertible**[cite: 1]. Constant inputs $x_1(t) = 0 and $x_2(t) = 5$ both yield a derivative of $y(t) = 0$[cite: 1].
- **(k)** $y[n] = x[n+1]$ for n \ge 0$; $y[n] = x[n] for n \le -1$[cite: 1]. **Not invertible**[cite: 1]. The value of $x[0] is completely omitted from the output sequence[cite: 1]. Therefore, $x_1[n] = 0$ and $x_2[n] = \delta[n]$ both result in $y[n] = 0$ for all $n$[cite: 1].
- **(l)** $y(t) = x(2t)$[cite: 1]. **Invertible**[cite: 1]. The inverse system represents a time stretch: $x(t) = y(t/2)$[cite: 1].
- **(m)** y[n] = x[2n]$[cite: 1]. **Not invertible**[cite: 1]. The system discards all odd-indexed values of the input sequence[cite: 1]. Inputs $x_1[n] = 0 and $x_2[n] =$

$\delta[n-1]$ both yield $y[n] = 0$[cite: 1].

- **(n)** $y[n] = x[n/2]$ for $n$ even; $0$ for $n$ odd[cite: 1]. **Invertible**[cite: 1]. The inverse system is a decimation operation: $x[n] = y[2n]$[cite: 1].

*23/08/2026, 10:40:18*

Determine the source of these questions as I want to rpadctice much more quesetion like thetese so I want to know the derivated state of tehse questions like where are they derived from?

*23/08/2026, 12:07:23*

**Thinking steps**

**Identifying the Source**

I've clarified the user's need; source identification and similar practice questions are the focus.

**Pinpointing the Resource**

The precise problems now link directly to *Signals and Systems*, Chapter 1, Oppenheim, Willsky, and Nawab, 2nd edition.

**Refining the Output**

I've decided to give a standard, generic response without using any user data, and identified the book and chapter.

✓ Done

These questions are derived directly from one of the most universally recognized textbooks for this subject.

They are the end-of-chapter problems from **Chapter 1** of:

- **Title:** *Signals and Systems*
- **Authors:** Alan V. Oppenheim, Alan S. Willsky, with S. Hamid Nawab
- **Edition:** 2nd Edition

The numbering in your images (1.16 through 1.30) perfectly matches the numbering in the "Problems" section of Chapter 1 in this book.

### Where to Practice More

If you want to practice more questions of this exact style and difficulty:

1. **Get the Textbook:** The rest of the chapters in Oppenheim and Willsky (especially Chapters 2 and 3 covering LTI systems, Fourier Series, and Fourier Transforms) are filled with hundreds of similar conceptual and mathematical problems.
2. **Other Standard Texts:** If you exhaust Oppenheim and want different perspectives or additional problems, the other gold-standard textbooks in this field are:
   - *Signals and Systems* by Simon Haykin and Barry Van Veen.
   - *Linear Systems and Signals* by B.P. Lathi.
   - *Schaum's Outline of Signals and Systems* by Hwei P. Hsu (This is highly recommended if your primary goal is just grinding through practice problems, as it contains hundreds of fully solved examples).

*23/08/2026, 12:07:23*

SOlve 1.17 b part using pricniple of additivity and homogenouity like the one requreid for linearlity like?

23/08/2026, 12:33:56

**Thinking steps**

**Defining Linearity Verification**

I've established the system and problem, focusing on additivity and scaling to test for linearity, to solve the problem.

**Verifying Linearity Proof**

I've formulated a complete, refined proof, including additivity, homogeneity, and conclusion, now ready for LaTeX.

Done

To rigorously prove if a system is linear, we must independently verify two properties: **Additivity** and **Homogeneity** (Scaling).

The continuous-time system is defined by the transformation $\mathcal{T}$:

$$y(t) = \mathcal{T}\{x(t)\} = x(\sin(t))$$

### 1. Proof of Additivity

The additivity principle states that the response to a sum of inputs must equal the sum of the responses to each individual input.

- Let a first input signal $x_1(t)$ yield the output $y_1(t)$:

$$y_1(t) = x_1(\sin(t))$$

- Let a second input signal $x_2(t)$ yield the output $y_2(t)$:

$$y_2(t) = x_2(\sin(t))$$

- Now, apply a new input $x_3(t)$ which is the sum of the two inputs:

$$x_3(t) = x_1(t) + x_2(t)$$

- The system output for $x_3(t)$ will be:

$$y_3(t) = \mathcal{T}\{x_3(t)\} = x_3(\sin(t))$$

- Substituting the definition of $x_3(t)$ into the equation:

$$y_3(t) = x_1(\sin(t)) + x_2(\sin(t))$$

- Comparing this to our initial outputs, we see:

$$y_3(t) = y_1(t) + y_2(t)$$

Since $\mathcal{T}\{x_1(t) + x_2(t)\} = \mathcal{T}\{x_1(t)\} + \mathcal{T}\{x_2(t)\}$, the system **satisfies the additivity property**.

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## 2. Proof of Homogeneity (Scaling)

The homogeneity principle states that scaling the input by a constant $a$ must scale the output by that exact same constant $a$.

- Let an input signal $x_1(t)$ yield the output $y_1(t)$:

$$y_1(t) = x_1(\sin(t))$$

- Now, apply a new input $x_2(t)$ which is the original input scaled by a constant $a$:

$$x_2(t) = a \cdot x_1(t)$$

- The system output for $x_2(t)$ will be:

$$y_2(t) = \mathcal{T}\{x_2(t)\} = x_2(\sin(t))$$

- Substituting the definition of $x_2(t)$ into the equation:

$$y_2(t) = a \cdot x_1(\sin(t))$$

- Comparing this to our initial output, we see:

$$y_2(t) = a \cdot y_1(t)$$

Since $\mathcal{T}\{a \cdot x_1(t)\} = a \cdot \mathcal{T}\{x_1(t)\}$, the system **satisfies the homogeneity property**.

## Conclusion

Because the system $y(t) = x(\sin(t))$ independently satisfies both additivity and homogeneity, **the system is linear.**

*23/08/2026, 12:33:56*

Solve for 1.18 a part and 1.19 one also like the above done in detailed

*23/08/2026, 12:50:17*

**Thinking steps**

**Initiating Problem Analysis**

I've initiated a deep dive into problem requirements, focusing on additivity and homogeneity principles for rigorous solutions.

**Refining Linearity Proofs**

I have formalized linearity proofs for both problems using additivity and homogeneity rigorously.

**Analyzing System Properties**

I've tested linearity and time-invariance across three systems using rigorous additivity/homogeneity and delay methods.

✓ Done

Here are the rigorous step-by-step proofs for determining linearity (via additivity and homogeneity) and time-invariance for these problems.

## Problem 1.18 (a): First Question (Linearity)

The discrete-time system is defined by the transformation $\mathcal{T}$:
$$y[n] = \mathcal{T}{x[n]} = \sum_{k=n-n_0}^{n+n_0} x[k]$$[cite: 1]

### 1. Proof of Additivity

- Let inputs $x_1[n]$ and $x_2[n]$ produce outputs $y_1[n]$ and $y_2[n]$[cite: 1]:
$$y_1[n] = \sum_{k=n-n_0}^{n+n_0} x_1[k], \quad y_2[n] = \sum_{k=n-n_0}^{n+n_0} x_2[k]$$[cite: 1]
- Apply the sum of the inputs: $x_3[n] = x_1[n] + x_2[n]$[cite: 1].
- The system output for $x_3[n]$ is:
$$y_3[n] = \sum_{k=n-n_0}^{n+n_0} (x_1[k] + x_2[k])$$[cite: 1]

- Because summation is distributive, we can split this: $$y_3[n] = \sum_{k=n-n_0}^{n+n_0} x_1[k] + \sum_{k=n-n_0}^{n+n_0} x_2[k] = y_1[n] + y_2[n]$$[cite: 1]
- **Additivity holds.**

### 2. Proof of Homogeneity

- Apply a scaled input: $x_4[n] = a \cdot x_1[n]$[cite: 1].
- The system output for $x_4[n]$ is: $$y_4[n] = \sum_{k=n-n_0}^{n+n_0} (a \cdot x_1[k])$$[cite: 1]
- Constants can be pulled out of a summation: $$y_4[n] = a \sum_{k=n-n_0}^{n+n_0} x_1[k] = a \cdot y_1[n]$$[cite: 1]
- **Homogeneity holds.**

**Conclusion:** Since both properties hold, **the system is linear**[cite: 1].

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## Problem 1.19: Linearity and Time-Invariance

To prove time-invariance, we must check if delaying the input by $t_0$ (or N_0$) yields the exact same result as delaying the original output by $t_0 (or N_0$). Let the delay operator be $\mathcal{D}.

### (a) $y(t) = t^2 x(t - 1)$[cite: 1]

**Linearity:**

- **Additivity:** $\mathcal{T}{x_1(t) + x_2(t)} = t^2 [x_1(t - 1) + x_2(t - 1)] = t^2 x_1(t - 1) + t^2 x_2(t - 1) = y_1(t) + y_2(t)$[cite: 1].
- **Homogeneity:** $\mathcal{T}{a \cdot x_1(t)} = t^2 [a \cdot x_1(t - 1)] = a \cdot [t^2 x_1(t - 1)] = a \cdot y_1(t)$[cite: 1].
- **Result: Linear.**

**Time-Invariance:**

- Delay the input by $t_0$: Let $x_2(t) = x_1(t - t_0)$[cite: 1]. The output is $y_{input_delayed}(t) = t^2 x_2(t - 1) = t^2 x_1(t - 1 - t_0)$[cite: 1].

- Delay the original output by $t_0$: $y_{output_delayed}(t) = y_1(t - t_0) = (t - t_0)^2 x_1((t - t_0) - 1)$[cite: 1].
- Comparing the two: $t^2 x_1(t - t_0 - 1) \neq (t - t_0)^2 x_1(t - t_0 - 1)$[cite: 1].
- **Result: Time-varying.**

### (b) $y[n] = x^2[n - 2]$[cite: 1]

**Linearity:**

- **Homogeneity Check:** $\mathcal{T}{a \cdot x_1[n]} = (a \cdot x_1[n - 2])^2 = a^2 \cdot x_1^2[n - 2] = a^2 \cdot y_1[n]$[cite: 1].
- Because $a^2 \cdot y_1[n] \neq a \cdot y_1[n]$ (unless $a = 1$ or $a=0$), the homogeneity property fails[cite: 1]. Additivity also fails: $(x_1 + x_2)^2 \neq x_1^2 + x_2^2$[cite: 1].
- **Result: Non-linear.**

**Time-Invariance:**

- Delay the input by $N_0$: Let $x_2[n] = x_1[n - N_0]$[cite: 1]. The output is $y_{input_delayed}[n] = x_2^2[n - 2] = (x_1[n - 2 - N_0])^2$[cite: 1].
- Delay the original output by $N_0$: $y_{output_delayed}[n] = y_1[n - N_0] = (x_1[(n - N_0) - 2])^2$[cite: 1].
- Both expressions result in $(x_1[n - N_0 - 2])^2$[cite: 1].
- **Result: Time-invariant.**

### (c) $y[n] = x[n + 1] - x[n - 1]$[cite: 1]

**Linearity:**

- **Additivity:** $\mathcal{T}{x_1[n] + x_2[n]} = (x_1[n + 1] + x_2[n + 1]) - (x_1[n - 1] + x_2[n - 1])$[cite: 1]. Grouping terms gives $(x_1[n + 1] - x_1[n - 1]) + (x_2[n + 1] - x_2[n - 1]) = y_1[n] + y_2[n]$[cite: 1].
- **Homogeneity:** $\mathcal{T}{a \cdot x_1[n]} = a \cdot x_1[n + 1] - a \cdot x_1[n - 1] = a(x_1[n + 1] - x_1[n - 1]) = a \cdot y_1[n]$[cite: 1].
- **Result: Linear.**

**Time-Invariance:**

- Delay the input by $N_0$: Let $x_2[n] = x_1[n - N_0]$[cite: 1]. The output is $y_{input_delayed}[n] = x_2[n + 1] - x_2[n - 1] = x_1[n + 1 - N_0] - x_1[n - 1 - N_0]$[cite: 1].
- Delay the original output by $N_0$: $y_{output_delayed}[n] = y_1[n - N_0] = x_1[(n - N_0) + 1] - x_1[(n - N_0) - 1]$[cite: 1].
- Both expressions match perfectly[cite: 1].
- **Result: Time-invariant.**

### (d) $y[n] = \mathcal{O}d{x[n]}$[cite: 1]

*The odd part of a signal is defined mathematically as* $\mathcal{O}d\{x[n]\} = \frac{x[n]-x[-n]}{2}$.

**Linearity:**

- **Additivity:** $\mathcal{T}{x_1[n] + x_2[n]} = \frac{(x_1[n] + x_2[n]) - (x_1[-n] + x_2[-n])}{2}$[cite: 1]. Rearranging gives $\frac{x_1[n] - x_1[-n]}{2} + \frac{x_2[n] - x_2[-n]}{2} = y_1[n] + y_2[n]$[cite: 1].
- **Homogeneity:** $\mathcal{T}{a \cdot x_1[n]} = \frac{a \cdot x_1[n] - a \cdot x_1[-n]}{2} = a \left( \frac{x_1[n] - x_1[-n]}{2} \right) = a \cdot y_1[n]$[cite: 1].
- **Result: Linear.**

**Time-Invariance:**

- Delay the input by $N_0$: Let $x_2[n] = x_1[n - N_0]$[cite: 1]. The output is $y_{input_delayed}[n] = \frac{x_2[n] - x_2[-n]}{2} = \frac{x_1[n - N_0] - x_1[-n - N_0]}{2}$[cite: 1].
- Delay the original output by $N_0$: $y_{output_delayed}[n] = y_1[n - N_0] = \frac{x_1[n - N_0] - x_1[-(n - N_0)]}{2} = \frac{x_1[n - N_0] - x_1[-n + N_0]}{2}$[cite: 1].
- Notice the signs on the $N_0$ term in the second half of the expressions: $-n - N_0 \neq -n + N_0$[cite: 1].
- **Result: Time-varying.**

Solve 1.18 b and c part in detailed

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**Thinking steps**

**Clarifying Problem Parameters**

I've precisely identified the user's need: solutions for 1.18 parts (b) and (c), focusing on time-invariance.

**Confirming Time-Invariance**

I've rigorously proven time-invariance by directly comparing shifted input with shifted output.

✓ Done

Here are the detailed, step-by-step proofs for parts (b) and (c) of Problem 1.18.

*(Note: In the original image, the second question asking "Is this system time-invariant?" is mislabeled with an "(a)". It will be treated as part (b) here.)*[cite: 1]

The discrete-time system is defined as:
$$y[n] = \sum_{k=n-n_0}^{n+n_0} x[k]$$[cite: 1]

---

### Part (b): Is this system time-invariant?

To prove that a system is time-invariant, we must show that shifting the input signal by an arbitrary time delay $N$ results in the exact same time shift in the output signal.

**Step 1: Determine the output when the input is delayed.**

- Let the original input be $x[n]$, which produces the output $y[n]$[cite: 1].
- Let's create a new input signal $x_1[n]$ that is delayed by $N$ samples: $x_1[n] = x[n-N]$[cite: 1].
- We feed this delayed input into the system to get a new output, $y_1[n]$:
  $$y_1[n] = \sum_{k=n-n_0}^{n+n_0} x_1[k]$$[cite: 1]
- Substitute $x_1[k]$ with $x[k-N]$:
  $$y_1[n] = \sum_{k=n-n_0}^{n+n_0} x[k-N]$$[cite: 1]

**Step 2: Determine the delayed version of the original output.**

- We take our original output equation and simply replace every instance of $n$ with $(n-N)$[cite: 1]:
  $$y[n-N] = \sum_{k=(n-N)-n_0}^{(n-N)+n_0} x[k]$$[cite: 1]

**Step 3: Compare $y_1[n]$ and $y[n-N]$.**

- Take the equation for $y_1[n]$ and introduce a change of variables for the summation index. Let $m = k - N$[cite: 1].
- We must also update the limits of the summation:
  - When $k = n - n_0$, the new lower limit is $m = (n - n_0) - N = (n - N) - n_0$[cite: 1].
  - When $k = n + n_0$, the new upper limit is $m = (n + n_0) - N = (n - N) + n_0$[cite: 1].
- Substituting $m$ into the $y_1[n]$ equation yields:
  $$y_1[n] = \sum_{m=(n-N)-n_0}^{(n-N)+n_0} x[m]$$[cite: 1]
- Notice that the dummy variable of summation ($m$ versus $k$) does not change the result of the sum. Therefore, this expression is mathematically identical to $y[n-N]$[cite: 1].

**Conclusion:**
Because $y_1[n] = y[n-N]$, **the system is time-invariant.**[cite: 1]

---

## Part (c): Express $C$ in terms of $B$ and $n_0$

We are given that the input signal is bounded by a finite integer $B$, meaning $|x[n]| < B$ for all n$[cite: 1]. We need to find the maximum possible bound $C for the output signal $y[n]$[cite: 1].

**Step 1: Set up the absolute value of the output.**
$$|y[n]| = \left| \sum_{k=n-n_0}^{n+n_0} x[k] \right|$$[cite: 1]

**Step 2: Apply the triangle inequality.**
The generalized triangle inequality states that the absolute value of a sum is less than or equal to the sum of the absolute values: $\left| \sum a_i \right| \le \sum |a_i|$[cite:

1].
$$|y[n]| \le \sum_{k=n-n_0}^{n+n_0} |x[k]|$$[cite: 1]

**Step 3: Apply the given input bound.**

We know that for every term in the sequence, |x[k]| < B$[cite: 1]. Therefore, if we replace every term in the sum with $B, the sum becomes strictly larger:
$$|y[n]| < \sum_{k=n-n_0}^{n+n_0} B$$[cite: 1]

**Step 4: Evaluate the summation.**

We are summing a constant value $B$ over a specific range of indices[cite: 1]. To find the total, we multiply $B$ by the total number of terms in the summation[cite: 1].

- The number of terms in a sum from index $A$ to index $Z$ is $(Z - A) + 1$[cite: 1].
- Here, upper limit $Z = n + n_0$[cite: 1].
- Lower limit $A = n - n_0$[cite: 1].
- Number of terms $= (n + n_0) - (n - n_0) + 1 = n + n_0 - n + n_0 + 1 = 2n_0 + 1$[cite: 1].

Multiplying the constant $B$ by the number of terms yields:
$$|y[n]| < (2n_0 + 1)B$$[cite: 1]

**Conclusion:**

The problem states that |y[n]| < C$[cite: 1]. Based on our derivation, the value of $C is $C = B(2n_0 + 1)$.[cite: 1]

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Do them with stating linear additivity property with constants a and b are they also needed here state the conditions for the above in detailed

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**Thinking steps**

**Clarifying the Core Task**

I've clarified the user's focus on the linear additivity property and its constants for the problem.

**Distinguishing System Properties**

I've separated and defined linearity, time-invariance, and stability to clarify distinctions in this analysis.

✓ Done

To answer your question directly: **No, the constants $a$ and $b$ are NOT needed for parts (b) and (c).**

It is very common to get these mixed up, so let's clarify exactly why. In signal processing, **Linearity**, **Time-Invariance**, and **Stability** are three completely independent properties. You must test them using different mathematical conditions.

Here is a breakdown of the exact conditions required for each property, followed by the detailed solutions for (b) and (c) using their correct conditions.

---

## The Three Independent System Conditions

### 1. Linearity Condition (Requires constants $a$ and $b$)

To prove a system is linear (which you did in part a), you must use the Superposition Principle. This combines Additivity and Homogeneity using arbitrary constants $a$ and $b$.

- **The Condition:** If $x_1[n] \to y_1[n]$ and $x_2[n] \to y_2[n]$, then the system must satisfy:

$$\mathcal{T}\{a \cdot x_1[n] + b \cdot x_2[n]\} = a \cdot y_1[n] + b \cdot y_2[n]$$

### 2. Time-Invariance Condition (No constants needed)

To prove a system is time-invariant (Part b), we do not scale or add signals. We only check if delaying the input causes an identical delay in the output.

- **The Condition:** If an input $x[n]$ produces output $y[n]$, then a delayed input $x[n-N]$ must produce exactly $y[n-N]$.

**3. Stability Condition / BIBO (No constants needed)**
To prove a system is stable (Part c), we only look at bounds (maximum possible values). We check Bounded-Input, Bounded-Output (BIBO).

- **The Condition:** If the input has a finite maximum bound ($|x[n]| \le B < \infty$), the output must also have a finite maximum bound ($|y[n]| \le C < \infty$).

---

## Detailed Solution for 1.18 (b): Time-Invariance

**System Definition:**
$$y[n] = \sum_{k=n-n_0}^{n+n_0} x[k]$$[cite: 1]

**Goal:** Test the Time-Invariance condition: Does $\mathcal{T}\{x[n-N]\} = y[n-N]$?

**Step 1: Find the output for a delayed input.**
Let's define a new input that is delayed by an integer $N$:
$$x_{delayed}[n] = x[n-N]$$[cite: 1]
Feed this into the system equation:
$$y_{delayed_input}[n] = \sum_{k=n-n_0}^{n+n_0} x_{delayed}[k]$$[cite: 1]
Substitute the definition of $x_{delayed}[k]$:
$$y_{delayed_input}[n] = \sum_{k=n-n_0}^{n+n_0} x[k-N]$$[cite: 1]

**Step 2: Find the delayed version of the original output.**
Take the original output equation $y[n]$ and replace every instance of the variable $n$ with $(n-N)$[cite: 1]:
$$y[n-N] = \sum_{k=(n-N)-n_0}^{(n-N)+n_0} x[k]$$[cite: 1]

**Step 3: Compare the two results.**
We need to see if $y_{delayed\_input}[n]$ and $y[n-N]$ are mathematically the same. Let's perform a change of variables on the summation index in Step 1.
Let $m = k - N$[cite: 1].
Now, we must adjust the summation limits for this new variable $m$:

- Lower limit: When $k = n - n_0$, $m = (n - n_0) - N = (n - N) - n_0$[cite: 1].

- Upper limit: When $k = n + n_0$, $m = (n + n_0) - N = (n - N) + n_0$[cite: 1].

Substitute $m$ and the new limits back into the equation from Step 1:
$$y_{delayed_input}[n] = \sum_{m=(n-N)-n_0}^{(n-N)+n_0} x[m]$$[cite: 1]

Because changing the dummy index letter from $m$ to $k$ doesn't change the mathematical sum, this is identical to the equation in Step 2[cite: 1].
Since $y_{delayed\_input}[n] = y[n - N]$, **the system satisfies the time-invariance condition.**[cite: 1]

---

## Detailed Solution for 1.18 (c): Stability (Finding bound $C$)

**System Definition:**
$$y[n] = \sum_{k=n-n_0}^{n+n_0} x[k]$$[cite: 1]

**Goal:** Apply the Stability condition. Given that $|x[n]| < B$, find the constant $C$ such that $|y[n]| < C$[cite: 1].

**Step 1: Take the absolute value of both sides of the system equation.**
$$|y[n]| = \left| \sum_{k=n-n_0}^{n+n_0} x[k] \right|$$[cite: 1]

**Step 2: Apply the Triangle Inequality.**
The absolute value of a sum is always less than or equal to the sum of the absolute values.
$$|y[n]| \le \sum_{k=n-n_0}^{n+n_0} |x[k]|$$[cite: 1]

**Step 3: Apply the input condition.**
We are given the condition that $|x[n]| < B$ for all n$[cite: 1]. We can replace the $|x[k]| term with its maximum bound $B$. Because $B$ is strictly greater than $|x[k]|$, the inequality changes to strictly less than:
$$|y[n]| < \sum_{k=n-n_0}^{n+n_0} B$$[cite: 1]

**Step 4: Solve the summation.**
We are summing a constant value ($B$) repeatedly[cite: 1]. The result is simply the constant $B$ multiplied by the total number of terms in the summation[cite: 1].

- The formula for the number of terms in an inclusive sequence from index $start$ to index $end$ is: $(end - start) + 1$[cite: 1].
- Number of terms $= (n + n_0) - (n - n_0) + 1$[cite: 1]
- Number of terms $= n + n_0 - n + n_0 + 1 = 2n_0 + 1$[cite: 1]

Multiply the constant $B$ by the number of terms:

$$|y[n]| < B(2n_0 + 1)$$[cite: 1]

Because the problem asks us to express the final bound as $C$[cite: 1], we conclude that $C = B(2n_0 + 1)$.[cite: 1]

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Solve 1.30 each part in detailed

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**Thinking steps**

**Defining the Invertibility**

I have clearly defined the goals and pinpointed invertibility as the core concept to be investigated for problem 1.30.

**Evaluating Invertibility**

I've rigorously analyzed each part of problem 1.30, applying the invertibility definition with examples.

**Determining Invertibility**

I have successfully analyzed systems (f) through (i), including deriving an inverse for several of them.

**Analyzing Invertibility**

I have successfully determined invertibility for (j), (k), (l), (m), and (n), including counterexamples or inverse functions.

✓ Done

Here are the detailed, step-by-step solutions for each part of problem 1.30 from "Assignment 2.docx"[cite: 1].

A system is **invertible** if distinct inputs always produce distinct outputs. If a system is invertible, we can mathematically rearrange the relationship to solve for $x$ in terms of $y$. If it is not invertible, we must prove it by providing a counterexample (two distinct input signals, $x_1$ and $x_2$, that yield the exact same output $y$).

---

### (a) $y(t) = x(t-4)$[cite: 1]

- **Is it invertible?** Yes. This system simply delays the signal by 4 time units. No information is lost.
- **Inverse System:** To recover the original input $x(t)$, we just need to advance the output $y(t)$ by 4 time units. Let $\tau = t - 4$, which means $t = \tau + 4$. Substituting this yields $y(\tau + 4) = x(\tau)$.

- Therefore, the inverse system is: $x(t) = y(t+4)$

### (b) $y(t) = \cos[x(t)]$[cite: 1]

- **Is it invertible?** No. The cosine function is periodic, meaning multiple different input values will produce the exact same output value.
- **Counterexample:**
  - Let $x_1(t) = 0$. The output is $y_1(t) = \cos(0) = 1$.
  - Let $x_2(t) = 2\pi$. The output is $y_2(t) = \cos(2\pi) = 1$.
  - Since $x_1(t) \neq x_2(t)$ but $y_1(t) = y_2(t)$, the system is not invertible.

### (c) $y[n] = nx[n]$[cite: 1]

- **Is it invertible?** No. At time index $n = 0$, the system multiplies the input by zero. Any information contained in $x[0]$ is permanently destroyed.
- **Counterexample:**
  - Let $x_1[n] = \delta[n]$ (an impulse of height 1 at $n=0$).
  - Let $x_2[n] = 2\delta[n]$ (an impulse of height 2 at $n=0$).
  - For both inputs, at $n = 0$, the output is $y[0] = 0 \cdot x[0] = 0$. For all other $n$, $y[n]$ is $0$ because the input is $0$. Both inputs yield $y[n] = 0$ everywhere.

### (d) $y(t) = \int_{-\infty}^{t} x(\tau) d\tau$[cite: 1]

- **Is it invertible?** Yes. This is an ideal integrator. Assuming the input function is reasonably well-behaved, integration can be reversed by differentiation.
- **Inverse System:** By applying the Fundamental Theorem of Calculus, taking the derivative of the output with respect to $t$ yields the input:
  - $x(t) = \frac{dy(t)}{dt}$

### (e) $y[n] = \begin{cases} x[n-1], & n \geq 1 \\ 0, & n = 0 \\ x[n], & n \leq -1 \end{cases}$ [cite: 1]

- **Is it invertible?** Yes. Let's trace where each input sample goes. For negative indices, $x$ is mapped directly to the same index in $y$ ($y[-1] = x[-1]$). For $n \geq 0$, the input $x[0]$ is stored in $y[1]$, $x[1]$ in $y[2]$, etc. Every sample of $x[n]$ is preserved in the output; the value at $y[0]$ is just a dummy zero we can ignore.
- **Inverse System:**
  - For $n \leq -1$: $x[n] = y[n]$
  - For $n \geq 0$: $x[n] = y[n+1]$

### (f) $y[n] = x[n]x[n-1]$[cite: 1]

- **Is it invertible?** No. Squaring or multiplying signal samples by themselves generally destroys sign information.
- **Counterexample:**
  - Let $x_1[n] = 1$ for all $n$ (a constant DC signal). The output is $y_1[n] = (1)(1) = 1$.
  - Let $x_2[n] = -1$ for all $n$. The output is $y_2[n] = (-1)(-1) = 1$.
  - Distinct inputs yield the same output.

### (g) $y[n] = x[1-n]$[cite: 1]

- **Is it invertible?** Yes. This is a time reversal accompanied by a time shift. Time reversals preserve all data points, they just flip the sequence axis.
- **Inverse System:** Let's define a new variable $m = 1 - n$. This means $n = 1 - m$. Substituting this back into the original equation gives $y[1-m] = x[m]$. Swapping the dummy variable $m$ back to $n$ gives the inverse:
  - $x[n] = y[1-n]$ (The system is its own inverse).

### (h) $y(t) = \int_{-\infty}^{t} e^{-(t-\tau)} x(\tau) d\tau$[cite: 1]

- **Is it invertible?** Yes. We can extract $x(t)$ by manipulating the equation and differentiating.
- **Inverse System:**
  1. First, pull the $e^{-t}$ term out of the integral since it doesn't depend on $\tau$: $y(t) = e^{-t} \int_{-\infty}^{t} e^{\tau} x(\tau) d\tau$

2. Multiply both sides by $e^t$: $e^t y(t) = \int_{-\infty}^{t} e^{\tau} x(\tau) d\tau$
3. Take the derivative of both sides with respect to $t$. Apply the product rule on the left, and the Fundamental Theorem of Calculus on the right:
   $e^t \frac{dy(t)}{dt} + e^t y(t) = e^t x(t)$
4. Divide entirely by $e^t$:
   $x(t) = \frac{dy(t)}{dt} + y(t)$

### (i) $y[n] = \sum_{k=-\infty}^{n} (1/2)^{n-k} x[k]$[cite: 1]

- **Is it invertible?** Yes. We can write this infinite sum recursively.
- **Inverse System:**

1. Expand the sum by peeling off the very last term (where $k=n$):
   $y[n] = (1/2)^0 x[n] + \sum_{k=-\infty}^{n-1} (1/2)^{n-k} x[k]$
2. Factor out a $(1/2)$ from the remaining summation:
   $y[n] = x[n] + \frac{1}{2} \sum_{k=-\infty}^{n-1} (1/2)^{n-1-k} x[k]$
3. Notice that the summation term is the exact definition of $y[n-1]$. Therefore:
   $y[n] = x[n] + \frac{1}{2} y[n-1]$
4. Rearrange to solve for $x[n]$:
   $x[n] = y[n] - \frac{1}{2} y[n-1]$

### (j) $y(t) = \frac{dx(t)}{dt}$[cite: 1]

- **Is it invertible?** No. The derivative of any constant (DC offset) is zero. The system permanently loses any baseline DC offset information of the input signal.
- **Counterexample:**
  - Let $x_1(t) = 0$. The output is $y_1(t) = \frac{d(0)}{dt} = 0$.
  - Let $x_2(t) = 5$. The output is $y_2(t) = \frac{d(5)}{dt} = 0$.
  - Distinct inputs yield the same output.

### (k) $y[n] = \begin{cases} x[n+1], & n \geq 0 \\ x[n], & n \leq -1 \end{cases}$[cite: 1]

- **Is it invertible?** No. Let's analyze where the input samples go.

- For $n \le -1$: $y[-1] = x[-1]$, $y[-2] = x[-2]$, etc.
- For $n \ge 0$: $y[0] = x[1]$, $y[1] = x[2]$, etc.
- Notice that the sample $x[0]$ is completely skipped and never appears in the output sequence $y[n]$.

- **Counterexample:**
  - Let $x_1[n] = 0$ for all $n$. Then $y_1[n] = 0$ everywhere.
  - Let $x_2[n] = \delta[n]$ (which is $1$ at $n = 0$, and $0$ elsewhere). Because $x[0]$ is ignored by the system, $y_2[n] = 0$ everywhere.
  - Distinct inputs yield the same output.

### (l) $y(t) = x(2t)$[cite: 1]

- **Is it invertible?** Yes. This system compresses the signal in time by a factor of 2. In continuous time, we can simply stretch it back out.
- **Inverse System:** Let $\tau = 2t$, which means $t = \tau/2$. Substituting this into the equation gives $y(\tau/2) = x(\tau)$. Swapping the dummy variable back to $t$ gives:
  - $x(t) = y(t/2)$

### (m) $y[n] = x[2n]$[cite: 1]

- **Is it invertible?** No. This is called *decimation* in discrete-time signal processing. It keeps only the even-indexed samples and discards all odd-indexed samples (e.g., $x[1], x[3], x[5]$ are lost forever).
- **Counterexample:**
  - Let $x_1[n] = 0$ for all $n$.
  - Let $x_2[n] = \delta[n-1]$ (a signal that is $1$ at $n = 1$, and $0$ elsewhere).
  - Since $x_2$ only has a non-zero value at an odd index, the decimation system discards it. Therefore, both $x_1$ and $x_2$ yield $y[n] = 0$ everywhere.

### (n) $y[n] = \begin{cases} x[n/2], & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$[cite: 1]

- **Is it invertible?** Yes. This is called *interpolation* (or upsampling). It takes the input sequence, spreads it out by placing it at the even indices, and stuffs zeros into the odd indices. No data is lost; it is simply spaced out.
- **Inverse System:** To recover $x$, we just need to read the even-indexed values of $y$ and ignore the zeros at the odd indices.
  - Let $n = 2k$ (which forces $n$ to be even). Then $k = n/2$.
  - By the system definition, $y[2k] = x[2k/2] = x[k]$.
  - Changing the index variable back to $n$, the inverse system is a decimation:
  - $x[n] = y[2n]$

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(d) $y(t) = \int_{-\infty}^{t} x(\tau)d\tau$

DOCX

- **Is it invertible?** Yes. This is an ideal integrator. Assuming the input function is reasonably well-behaved, integration can be reversed by differentiation.
- **Inverse System:** By applying the Fundamental Theorem of Calculus, taking the derivative of the output with respect to $t$ yields the input:
    - $x(t) = \frac{dy(t)}{dt}$

Continue the derivation as the queston states for the inverse system also

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**Thinking steps**

**Defining the Inverse System**

I have successfully defined the inverse system as the derivative, confirming it's invertible as per the prior results.

**Deriving the Inverse System**

I've rigorously derived the inverse for part (d) using the Fundamental Theorem of Calculus, addressing the user's specific request.

✓ Done

Here is the rigorous mathematical derivation to find the inverse system for part (d) shown in "image_65eee4.png".

The system is defined as:

$$y(t) = \int_{-\infty}^{t} x(\tau)d\tau$$

To construct the inverse system, we must algebraically isolate and solve for the input signal, $x(t)$, in terms of the output signal, $y(t)$.

**Step 1: Differentiate both sides**

We apply the derivative with respect to time ($t$) to both sides of the equation:

$$\frac{dy(t)}{dt} = \frac{d}{dt}\left[\int_{-\infty}^{t} x(\tau)d\tau\right]$$

**Step 2: Apply Leibniz's Rule (Fundamental Theorem of Calculus)**

To evaluate the derivative of the integral on the right side, we use the Leibniz Integral

Rule. The rule for differentiating a definite integral with variable limits is:

$$\frac{d}{dt}\left[\int_{a(t)}^{b(t)} f(\tau)d\tau\right] = f(b(t)) \cdot \frac{db(t)}{dt} - f(a(t)) \cdot \frac{da(t)}{dt}$$

For our specific system, the parameters map as follows:

- The integrand is $f(\tau) = x(\tau)$
- The upper limit is $b(t) = t$
- The lower limit is a constant $a(t) = -\infty$

**Step 3: Evaluate the terms**

Substitute our specific limits into the Leibniz formula:

$$\frac{d}{dt}\left[\int_{-\infty}^{t} x(\tau)d\tau\right] = x(t) \cdot \left(\frac{d}{dt}(t)\right) - \lim_{a \to -\infty}\left[x(a) \cdot \left(\frac{d}{dt}(a)\right)\right]$$

Now, we evaluate the derivatives of the limits:

- The derivative of the upper limit $t$ with respect to $t$ is $1$.
- The derivative of the constant lower limit $-\infty$ with respect to $t$ is $0$.

Plugging these back in:

$$\frac{d}{dt}\left[\int_{-\infty}^{t} x(\tau)d\tau\right] = x(t) \cdot (1) - x(-\infty) \cdot (0)$$

$$\frac{d}{dt}\left[\int_{-\infty}^{t} x(\tau)d\tau\right] = x(t)$$

**Step 4: State the final inverse system**

Substituting this result back into our equation from Step 1 gives the final inverse system:

$$x(t) = \frac{dy(t)}{dt}$$

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Solve e part in detailed

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**Thinking steps**

**Defining the Problem**

I've precisely defined the target problem, identifying the specific equation needed for 1.30 (e).

**Inverting the System**

I have successfully proven the system's invertibility and formulated its inverse equation.

Done

Here is the detailed, step-by-step derivation for part (e) of problem 1.30.

The discrete-time system is given as a piecewise function:
$$y[n] = \begin{cases} x[n-1], & n \ge 1 \ 0, & n = 0 \ x[n], & n \le -1 \end{cases}$$[cite: 1]

To determine if the system is invertible, we must check if any information from the input signal $x[n]$ is lost or overwritten[cite: 1]. We do this by mapping the input indices to the output indices.

**Step 1: Analyze the index mapping (Where does each $x[n]$ go?)**
Let's explicitly write out a few terms for each part of the piecewise function to see how the input $x$ is mapped to the output $y$[cite: 1].

- **For $n \leq -1$**: The rule is $y[n] = x[n]$[cite: 1].
  - $y[-1] = x[-1]$
  - $y[-2] = x[-2]$
  - $y[-3] = x[-3]$, and so on.
    *(Every input sample for negative time indices is stored at the exact same index in the output).*

- **For $n = 0$**: The rule is $y[0] = 0$[cite: 1].
  *(A dummy zero is inserted at index 0. We must ensure $x[0]$ wasn't destroyed).*

- **For $n \geq 1$**: The rule is $y[n] = x[n-1]$[cite: 1].
  - If $n = 1$: $y[1] = x[1-1] = x[0]$
  - If $n = 2$: $y[2] = x[2-1] = x[1]$
  - If $n = 3$: $y[3] = x[3-1] = x[2]$, and so on.
    *(Every input sample for non-negative time indices is delayed by one step and stored safely in the output).*

**Step 2: Conclusion on Invertibility**
Because every single input sample $x[k]$ (for all integers $k$ from $-\infty$ to \infty$) is uniquely mapped to a location in the output sequence $y[n], distinct inputs will always produce distinct outputs. No information is lost[cite: 1].

- **Result: Yes, the system is invertible.**

**Step 3: Construct the Inverse System**
To build the inverse system, we must write $x[n]$ as a function of $y[n]$. We will solve this case-by-case based on the index ranges of the input signal $x$. Let's use $k$ as our target input index.

- **Case 1: Recovering $x[k]$ for $k \leq -1$**
  From our original system, we know that for $n \leq -1$, $y[n] = x[n]$[cite: 1].
  Replacing $n$ with $k$, we get $y[k] = x[k]$.
  Therefore: $x[k] = y[k]$ **for** $k \leq -1$.

- **Case 2: Recovering $x[k]$ for $k \geq 0$**
  From our original system, we know that for $n \geq 1$, $y[n] = x[n-1]$[cite: 1].
  We want to find an expression for $x[k]$. Let $k = n - 1$.
  If $n \geq 1$, then $k$ must be $\geq 0$.
  Rearranging $k = n - 1$ gives us $n = k + 1$.
  Substitute $n = k + 1$ into the original equation: $y[k+1] = x[k]$.
  Therefore: $x[k] = y[k+1]$ **for** $k \geq 0$.

*(Note: The value $y[0] = 0$ produced by the forward system is simply ignored by the inverse system, as it contains no information about $x$).*[cite: 1]

**Step 4: State the final inverse system**

Combining the two cases and replacing the dummy variable $k$ back to standard time index $n$, we get the piecewise inverse system:

$$x[n] = \begin{cases} y[n+1], & n \geq 0 \\ y[n], & n \leq -1 \end{cases}$$

*23/08/2026, 17:46:57*

Solve f in detailed

23/08/2026, 18:00:31

**Thinking steps**

**Defining the Problem**

I've precisely defined the inversion problem, referencing the assignment and system equation.

**Proving Non-Invertibility**

I've proven the system non-invertible using multiple counterexamples, explicitly demonstrating information loss.

✓ Done

Here is the detailed, step-by-step solution for part (f) of problem 1.30.

The discrete-time system is defined as:
$$y[n] = x[n]x[n-1]$$[cite: 1]

**Step 1: Understand the condition for invertibility**
A system is invertible if and only if distinct input signals *always* produce distinct output signals. This guarantees a one-to-one mapping, allowing us to uniquely recover the input $x[n]$ just by looking at the output $y[n]$[cite: 1].

If we can find even one pair of different input signals that result in the exact same output signal, the system is **not invertible**[cite: 1].

**Step 2: Analyze the mathematical operation**
The system multiplies the current input sample $x[n]$ by the previous input sample $x[n-1]$[cite: 1]. There are two major mathematical red flags here that suggest information is lost:

1. **Multiplication by zero:** If $x[n-1] = 0$, then $y[n]$ will be $0$ regardless of what the current input $x[n]$ is. The value of $x[n]$ is permanently erased.
2. **Sign cancellation:** The product of two positive numbers yields a positive result, and the product of two negative numbers also yields a positive result. This suggests that absolute sign information might be lost.

**Step 3: Construct a counterexample**

To formally prove the system is not invertible, we will use the "sign cancellation" weakness to create two different inputs that produce the same output[cite: 1].

- **Input Signal 1:** Let $x_1[n] = 1$ for all values of $n$ (a constant DC signal of positive 1)[cite: 1].
  - Substitute this into the system: $y_1[n] = x_1[n] \cdot x_1[n-1]$[cite: 1].
  - Since the signal is constant, $x_1[n] = 1$ and $x_1[n-1] = 1$[cite: 1].
  - $y_1[n] = (1)(1) = 1$ for all $n$[cite: 1].
- **Input Signal 2:** Let $x_2[n] = -1$ for all values of $n$ (a constant DC signal of negative 1)[cite: 1].
  - Substitute this into the system: $y_2[n] = x_2[n] \cdot x_2[n-1]$[cite: 1].
  - Since the signal is constant, $x_2[n] = -1$ and $x_2[n-1] = -1$[cite: 1].
  - $y_2[n] = (-1)(-1) = 1$ for all $n$[cite: 1].

**Step 4: State the conclusion**

We have demonstrated that the input signal $x_1[n] = 1$ and the completely different input signal $x_2[n] = -1$ both yield the exact same output signal, $y[n] = 1$[cite: 1].

Because multiple inputs map to the same output, an inverse system cannot exist (it wouldn't know whether to output $1$ or $-1$). Therefore, **the system is not invertible.**[cite: 1]

*23/08/2026, 18:00:31*

Solve g in detailed

*23/08/2026, 18:00:58*

**Thinking steps**

**Defining the Problem**

I've clarified the user's need: a solution to problem 1.30(g), focusing on invertibility and the associated system analysis.

✓ Done

Here is the detailed, step-by-step solution for part (g) of problem 1.30.

The discrete-time system is defined as:
$$y[n] = x[1 - n]$$[cite: 1]

**Step 1: Analyze the system's operation**
To determine if a system is invertible, we must check if any input data is lost or if multiple inputs map to the same output. Let's look at how the indices of the input signal $x$ map to the indices of the output signal $y$[cite: 1]:

- When $n = -1$, the output is $y[-1] = x[1 - (-1)] = x[2]$
- When $n = 0$, the output is $y[0] = x[1 - 0] = x[1]$
- When $n = 1$, the output is $y[1] = x[1 - 1] = x[0]$
- When $n = 2$, the output is $y[2] = x[1 - 2] = x[-1]$

The operation $1 - n$ represents a time reversal (the $-n$ part) combined with a time shift. Because every integer $n$ produces a unique integer $1 - n$, every single sample of the input sequence is simply moved to a new, unique location in the output sequence. No samples are dropped, multiplied by zero, or merged together.

**Step 2: State the conclusion on invertibility**
Since there is a perfect one-to-one mapping between the input samples and the output samples, distinct inputs will always yield distinct outputs.

- **Result: Yes, the system is invertible.**[cite: 1]

**Step 3: Construct the Inverse System**

To build the inverse system, we need to mathematically rearrange the equation to express the original input $x$ as a function of the output $y$.

1. Start with the given system equation:
   $$y[n] = x[1 - n]$$[cite: 1]
2. Let's introduce a dummy index variable $k$ for the input signal. Let $k = 1 - n$.
3. We need to express $n$ in terms of $k$. Rearranging the equation gives:
   $$n = 1 - k$$
4. Now, substitute this expression for $n$ back into the left side of the original equation:
   $$y[1 - k] = x[k]$$
5. Rearrange the sides so the input $x$ is isolated on the left:
   $$x[k] = y[1 - k]$$
6. Finally, replace the dummy variable $k$ with the standard discrete-time index $n$:
   $$x[n] = y[1 - n]$$

**Step 4: Final Answer**

The inverse system is exactly the same as the forward system:

$$x[n] = y[1 - n]$$

[cite: 1]

*(Note: A system that is its own inverse is called an involution. If you apply the system twice, you get the original signal back: $y[1-n] = x[1-(1-n)] = x[n]$.)*

*23/08/2026, 18:00:58*

Solve h in detailed

23/08/2026, 18:02:30

**Thinking steps**

**Defining the Problem**

I've successfully identified and formulated the core mathematical problem to solve: the convolution integral.

**Deriving the Inverse**

I've successfully derived the inverse system through step-by-step differentiation and algebraic manipulation.

Done

Here is the detailed, step-by-step mathematical derivation for part (h) of problem 1.30.

The continuous-time system is defined as:

$$y(t) = \int_{-\infty}^{t} e^{-(t - \tau)} x(\tau) d\tau$$[cite: 1]

**Step 1: Understand the approach**

To prove the system is invertible and find its inverse, we need to mathematically isolate the input signal $x(t)$ in terms of the output signal $y(t)$. Because the input signal is trapped inside an integral, we will use differentiation to extract it, similar to an ideal integrator.

**Step 2: Simplify the integral**

The exponential term inside the integral is $e^{-(t-\tau)}$, which can be algebraically split into $e^{-t} \cdot e^{\tau}$[cite: 1].

Because the integration variable is $\tau$, the $e^{-t}$ term acts as a constant with respect to the integral. We can factor it out:

$$y(t) = e^{-t} \int_{-\infty}^{t} e^{\tau} x(\tau) d\tau$$[cite: 1]

**Step 3: Isolate the integral**

To make differentiation easier, multiply both sides of the equation by $e^{t}$ to move the exponential term to the left side:

$$e^{t} y(t) = \int_{-\infty}^{t} e^{\tau} x(\tau) d\tau$$[cite: 1]

**Step 4: Differentiate both sides**

Now, take the derivative of both sides with respect to time ($t$)[cite: 1].

- **Left Side (Product Rule):** The left side is a product of two functions of $t$ ($e^tand $y(t)$). Applying the product rule $\frac{d}{dt}[u \cdot v] = u \cdot \frac{dv}{dt} + v \cdot \frac{du}{dt} yields: $$\frac{d}{dt}[e^{t} y(t)] = e^{t} \frac{dy(t)}{dt} + y(t) \cdot e^{t}$$[cite: 1]

- **Right Side (Fundamental Theorem of Calculus):** Using the Leibniz integral rule (or Fundamental Theorem of Calculus), taking the derivative of an integral with respect to its upper limit simply evaluates the integrand at that upper limit: $$\frac{d}{dt} \left[ \int_{-\infty}^{t} e^{\tau} x(\tau) d\tau \right] = e^{t} x(t)$$[cite: 1]

Equating the results of the left and right sides gives: $$e^{t} \frac{dy(t)}{dt} + e^{t} y(t) = e^{t} x(t)$$[cite: 1]

**Step 5: Solve for $x(t)$**

To completely isolate $x(t)$, divide the entire equation by $e^t$ (since $e^t$ is never equal to zero, this is mathematically valid)[cite: 1]: $$\frac{dy(t)}{dt} + y(t) = x(t)$$[cite: 1]

**Conclusion:**

Because we successfully isolated $x(t)$ into a unique expression, the system does not lose information and **is invertible.**[cite: 1]

The final inverse system is:

$$x(t) = \frac{dy(t)}{dt} + y(t)$$

[cite: 1]

*23/08/2026, 18:02:30*

Solve  (i) in detailed

*23/08/2026, 18:02:51*

**Thinking steps**

**Defining the Problem**

I've clarified the user's need: a solution to problem 1.30(i), specifically determining and deriving its inverse if it exists.

**Deriving the Inverse**

I've successfully derived the inverse system equation: x[n] = y[n] - 0.5y[n-1], proving its invertibility.

✓ Done

Here is the detailed, step-by-step mathematical derivation for part (i) of problem 1.30.

The discrete-time system is defined as:
$$y[n] = \sum_{k=-\infty}^{n} \left(\frac{1}{2}\right)^{n-k} x[k]$$[cite: 1]

**Step 1: Understand the approach**
To determine if the system is invertible, we need to see if we can algebraically isolate $x[n]$ in terms of $y[n]$ and its delayed versions[cite: 1]. When dealing with infinite summations that have exponential terms, a standard technique is to write out the equation for $y[n]$, extract the most recent term, and find a recursive relationship with $y[n-1]$.

**Step 2: Expand the summation for $y[n]$**
Let's separate the summation into two parts: the very last term (where k = n$) and all the previous terms (from $k = -\infty up to $n - 1$).

$$y[n] = \left(\left(\frac{1}{2}\right)^{n-n} x[n]\right) + \sum_{k=-\infty}^{n-1} \left(\frac{1}{2}\right)^{n-k} x[k]$$

Since $(1/2)^0 = 1$, the first term simplifies to just $x[n]$:

$$y[n] = x[n] + \sum_{k=-\infty}^{n-1} \left(\frac{1}{2}\right)^{n-k} x[k]$$

**Step 3: Relate the remaining sum to $y[n-1]$**

We want to make the remaining summation look exactly like the definition of the system, but shifted by one time step.

First, let's write out what $y[n-1]$ should look like based on the original system definition by replacing every $n$ with $n-1$:

$$y[n-1] = \sum_{k=-\infty}^{n-1} \left(\frac{1}{2}\right)^{(n-1)-k} x[k]$$

Now, look back at the summation inside our expanded $y[n]$ equation. We can factor out a $(1/2)^1$ from the exponent:

$$\sum_{k=-\infty}^{n-1} \left(\frac{1}{2}\right)^{n-k} x[k] = \sum_{k=-\infty}^{n-1} \left(\frac{1}{2}\right)^{1} \cdot \left(\frac{1}{2}\right)^{n-1-k} x[k]$$

Pull the constant factor of $1/2$ outside the summation:

$$= \frac{1}{2} \sum_{k=-\infty}^{n-1} \left(\frac{1}{2}\right)^{n-1-k} x[k]$$

Notice that the remaining summation is now the exact definition of $y[n-1]$.

Therefore, we can substitute it:

$$= \frac{1}{2} y[n-1]$$

**Step 4: Solve for $x[n]$**

Substitute this result back into our expanded equation from Step 2:

$$y[n] = x[n] + \frac{1}{2} y[n-1]$$

Now, simply rearrange the equation to isolate $x[n]$:

$$x[n] = y[n] - \frac{1}{2} y[n-1]$$

**Conclusion:**

Because we can uniquely determine the current input sample $x[n]$ using only the current and previous output samples, no information is lost. **The system is invertible.**[cite: 1]

The final inverse system is:

$$x[n] = y[n] - \frac{1}{2}y[n-1]$$

[cite: 1]

*23/08/2026, 18:02:51*

Solve j in detailed

*23/08/2026, 18:03:17*

**Thinking steps**

**Defining the Problem**

I've successfully identified the core challenge: inverting or demonstrating the non-invertibility of system (j), y(t) = dx(t)/dt.

Done

Here is the detailed, step-by-step solution for part (j) of problem 1.30.

The continuous-time system is defined as:
$$y(t) = \frac{dx(t)}{dt}$$[cite: 1]

**Step 1: Understand the condition for invertibility**
A system is invertible if and only if distinct input signals *always* produce distinct output signals. This ensures a one-to-one mapping, meaning you can uniquely determine the exact input signal just by looking at the output signal.

To prove a system is **not invertible**, we only need to provide a single counterexample: two distinctly different input signals ($x_1$ and $x_2$) that result in the exact same output signal ($y$)[cite: 1].

**Step 2: Analyze the mathematical operation**
The system is a differentiator, meaning it calculates the instantaneous rate of change of the input signal with respect to time[cite: 1].

A fundamental property of calculus is that the derivative of any constant value is zero. Because of this, if a constant DC offset is added to an input signal, the differentiator will completely erase that constant. The system permanently loses information about the signal's baseline or starting value (the "initial condition").

**Step 3: Construct a counterexample**
We can use the property of constants to easily construct two different input signals that yield the same output[cite: 1].

- **Input Signal 1:** Let $x_1(t) = 0$ for all $t$ (a constant signal at zero).

    - Substitute this into the system: $y_1(t) = \frac{d}{dt}(0)$
    - The output is $y_1(t) = 0$.

- **Input Signal 2:** Let $x_2(t) = 5$ for all $t$ (a constant signal at amplitude five).
    - Substitute this into the system: $y_2(t) = \frac{d}{dt}(5)$
    - The output is $y_2(t) = 0$.

*(Note: You could also use moving signals, such as $x_1(t) = t$ and $x_2(t) = t + 5$, which both yield an output of $y(t) = 1$. Constant signals are just the simplest way to demonstrate the proof.)*

**Step 4: State the conclusion**

We have demonstrated that the input signal $x_1(t) = 0$ and the completely different input signal $x_2(t) = 5$ both yield the exact same output signal, $y(t) = 0$[cite: 1].

Because multiple different inputs map to the same output, it is impossible for an inverse system to determine which constant was originally put in. Therefore, **the system is not invertible.**[cite: 1]

*23/08/2026, 18:03:17*

solve k in detailed

*23/08/2026, 18:03:36*

**Thinking steps**

**Defining the Problem**

I've successfully identified and formulated the core problem: to determine the invertibility of the specified discrete-time system.

**Confirming Non-Invertibility**

I've proven the system's non-invertibility by constructing counterexamples showing distinct inputs result in identical outputs.

✓ Done

Here is the detailed, step-by-step solution for part (k) of problem 1.30.

The discrete-time system is defined as a piecewise function:
$$y[n] = \begin{cases} x[n+1], & n \ge 0 \ x[n], & n \le -1 \end{cases}$$[cite: 1]

**Step 1: Understand the condition for invertibility**
For a system to be invertible, it must preserve all information about the input signal. Every distinct input signal must produce a distinct output signal. If any part of the input signal is permanently lost or ignored, the system cannot be reversed, and it is **not invertible**[cite: 1].

To prove a system is not invertible, we must demonstrate that two different input signals yield the exact same output signal[cite: 1].

**Step 2: Analyze the index mapping (Where does each $x[n]$ go?)**
Let's explicitly write out a few terms for both parts of the piecewise function to see how the input samples $x$ map to the output sequence $y$[cite: 1].

- **For $n \le -1$**: The rule is $y[n] = x[n]$[cite: 1].
  - $y[-3] = x[-3]$
  - $y[-2] = x[-2]$
  - $y[-1] = x[-1]$
    *(For negative time, the output is exactly the same as the input).*

- **For $n \geq 0$:** The rule is $y[n] = x[n+1]$[cite: 1].
  - $y[0] = x[1]$
  - $y[1] = x[2]$
  - $y[2] = x[3]$
    *(For non-negative time, the output looks ahead and grabs the next input sample).*

**Step 3: Identify the lost information**
Look closely at the list of input samples ($x[-2], x[-1], x[1], x[2]$) that make it into the output signal. The sample $x[0]$ is completely skipped over.
Because $x[0]$ never appears anywhere in the output sequence $y[n]$, its value is permanently erased by the system.

**Step 4: Construct a counterexample**
Because the system ignores whatever happens at time $n = 0$, we can create two input signals that are identical everywhere *except* at $n=0$[cite: 1]. The system will produce the same output for both.

- **Input Signal 1:** Let $x_1[n] = 0$ for all $n$ (a sequence of all zeros).
  - Because the input is entirely zeros, the output is $y_1[n] = 0$ for all $n$.
- **Input Signal 2:** Let $x_2[n] = \delta[n]$ (the unit impulse, which is $1$ exactly at $n = 0$, and $0$ everywhere else).
  - For $n \leq -1$: $y_2[n] = x_2[n] = 0$.
  - For $n \geq 0$: $y_2[n] = x_2[n+1]$. Since $n \geq 0$, the index $(n+1)$ is always $\geq 1$. Because $\delta[n]$ is only non-zero at $n = 0$, $x_2[n+1] = 0$.
  - The output is $y_2[n] = 0$ for all $n$.

**Step 5: State the conclusion**
We have demonstrated that the completely empty input signal $x_1[n] = 0$ and the impulse signal $x_2[n] = \delta[n]$ both yield the exact same output signal, $y[n] = 0$[cite: 1].

Because multiple different inputs map to the same output, an inverse system cannot determine what the original value of $x[0]$ was. Therefore, **the system is not**

**invertible.**[cite: 1]

*23/08/2026, 18:03:36*

Solve I in detailed

23/08/2026, 18:04:06

**Thinking steps**

**Defining the Problem**

I've precisely defined the inversion problem, focusing on system invertibility and inverse construction for y(t) = x(2t).

Done

Here is the detailed, step-by-step solution for part (l) of problem 1.30.

The continuous-time system is defined as:
$$y(t) = x(2t)$$[cite: 1]

**Step 1: Understand the system's operation**
To determine if a system is invertible, we must check if any input information is lost or destroyed.
The operation $x(2t)$ represents a **time compression** in continuous time[cite: 1]. The system takes the original signal $x(t)$ and speeds it up by a factor of 2. For example, whatever value the input signal had at $t = 10$, the output signal will have that same value at $t = 5$.

Because time is continuous (not made of discrete, separated points), squeezing the signal does not discard any data points. Every single value of the input signal is perfectly preserved and simply mapped to a new, unique time instance in the output signal.

**Step 2: State the conclusion on invertibility**
Since there is a perfect one-to-one mapping between the time instances of the input and the time instances of the output, distinct inputs will always yield distinct outputs.

- **Result: Yes, the system is invertible.**[cite: 1]

**Step 3: Construct the Inverse System**
To build the inverse system, we must algebraically reverse the time scaling to express

the original input $x$ as a function of the output $y$. We do this by applying a substitution of variables.

1. Start with the given system equation:
   $$y(t) = x(2t)$$[cite: 1]
2. Let's introduce a new dummy time variable $\tau$ to represent the argument of the input function. Let $\tau = 2t$.
3. We need to express the output's time variable $t$ in terms of $\tau$. Rearranging the equation gives:
   $$t = \frac{\tau}{2}$$
4. Now, substitute these expressions for $t$ and $2t$ back into the original equation:
   $$y\left(\frac{\tau}{2}\right) = x(\tau)$$
5. Rearrange the sides so the input $x$ is isolated on the left:
   $$x(\tau) = y\left(\frac{\tau}{2}\right)$$
6. Finally, replace the dummy variable $\tau$ with the standard continuous-time variable $t$ :
   $$x(t) = y\left(\frac{t}{2}\right)$$

**Step 4: Final Answer**

The inverse system is a time-expansion (or time-stretching) operation that slows the signal back down by a factor of 2:

$$x(t) = y\left(\frac{t}{2}\right)$$

[cite: 1]

*23/08/2026, 18:04:06*

Solve m in detailed

*23/08/2026, 18:04:23*

**Thinking steps**

**Defining the Scope**

I have clearly defined the objective and identified the target problem within the source document.

✓ Done

Here is the detailed, step-by-step solution for part (m) of problem 1.30.

The discrete-time system is defined as:
$$y[n] = x[2n]$$[cite: 1]

**Step 1: Analyze the system's operation**
To determine if a system is invertible, we must check if any input information is lost or discarded. Let's map the indices of the input signal $x$ to the indices of the output signal $y$[cite: 1]:

- When $n = 0$, the output is $y[0] = x[0]$
- When $n = 1$, the output is $y[1] = x[2]$
- When $n = 2$, the output is $y[2] = x[4]$
- When $n = -1$, the output is $y[-1] = x[-2]$

This mathematical operation is called **decimation** (or downsampling) in discrete-time signal processing. It keeps the even-indexed samples of the input sequence and completely discards all the odd-indexed samples.

**Step 2: Identify the lost information**
Look at the input samples that actually make it to the output: $x[0], x[2], x[4], \dots$
The samples at odd indices—such as x[1], x[-1], x[3], x[-3]$—are skipped over and never appear in the output sequence $y[n]. Because these values are permanently erased by the system, we cannot uniquely reconstruct the original signal $x[n]$ just by looking at $y[n]$.

**Step 3: Construct a counterexample**

To formally prove a system is not invertible, we just need to provide two distinctly different input signals that produce the exact same output signal[cite: 1]. Since the system ignores all odd indices, we can create two signals that differ *only* at an odd index.

- **Input Signal 1:** Let $x_1[n] = 0$ for all $n$ (a sequence of all zeros).
  - Substitute into the system: $y_1[n] = x_1[2n]$.
  - Since the input is entirely zeros everywhere, the output is $y_1[n] = 0$ for all $n$.
- **Input Signal 2:** Let $x_2[n] = \delta[n-1]$ (a unit impulse that is $1$ exactly at $n = 1$, and $0$ everywhere else).
  - Substitute into the system: $y_2[n] = x_2[2n] = \delta[2n-1]$.
  - The impulse function $\delta[k]$ is only non-zero when its argument $k = 0$.
  - For the argument $(2n-1)$ to be equal to $0$, $n$ would have to be $0.5$. However, $n$ must be an integer in discrete time. Therefore, $(2n-1)$ is always an odd integer and is never equal to $0$.
  - This means the output is $y_2[n] = 0$ for all integer values of $n$.

**Step 4: State the conclusion**

We have demonstrated that the completely empty input signal $x_1[n] = 0$ and the impulse signal $x_2[n] = \delta[n-1]$ both yield the exact same output signal, $y[n] = 0$[cite: 1].

Because multiple different inputs map to the same output, an inverse system cannot determine what original values occupied the odd indices. Therefore, **the system is not invertible.**[cite: 1]

*23/08/2026, 18:04:23*

Solve n in detailed

23/08/2026, 18:04:44

**Thinking steps**

**Defining the Problem**

I've successfully identified the specific problem and located its relevant source image.

Done

Here is the detailed, step-by-step solution for part (n) of problem 1.30.

The discrete-time system is defined as a piecewise function:
$$y[n] = \begin{cases} x[n/2], & n \text{ even} \ 0, & n \text{ odd}
\end{cases}$$[cite: 1]

**Step 1: Analyze the system's operation**

To determine if a system is invertible, we must check if any input information is lost or destroyed. Let's explicitly map the indices of the input signal $x$ to the indices of the output signal $y$ to see what happens[cite: 1]:

- When $n = 0$ (even): $y[0] = x[0/2] = x[0]$
- When $n = 1$ (odd): $y[1] = 0$
- When $n = 2$ (even): $y[2] = x[2/2] = x[1]$
- When $n = 3$ (odd): $y[3] = 0$
- When $n = 4$ (even): $y[4] = x[4/2] = x[2]$
- When $n = -2$ (even): $y[-2] = x[-2/2] = x[-1]$

This operation is known as **interpolation** (or upsampling) in digital signal processing. The system takes the original input samples, spaces them out by placing them exclusively at the even indices of the output, and stuffs zeros into all the odd indices in between.

**Step 2: State the conclusion on invertibility**

Look closely at the mapping: every single sample of the input sequence $x[n]$ is perfectly preserved in the output sequence at an even index. No input samples are dropped, overwritten, or erased.

Because there is a perfect one-to-one mapping for every input data point, distinct input signals will always produce distinct output signals.

- **Result: Yes, the system is invertible.**[cite: 1]

**Step 3: Construct the Inverse System**

To build the inverse system, we must express the original input $x$ as a function of the output $y$. We need a mathematical operation that extracts the original samples and throws away the stuffed zeros.

1. Look at the rule for the even indices from the original system definition: For any even integer $n$, $y[n] = x[n/2]$[cite: 1].
2. Let's introduce a dummy variable $k$ to represent the index of the input sequence $x$ . We want to find $x[k]$.
3. From the relationship $y[n] = x[n/2]$, we can set $k = n/2$.
4. Rearranging this to solve for $n$ gives $n = 2k$. (Note that multiplying any integer $k$ by $2$ guarantees that $n$ is an even number, which perfectly aligns with our condition).
5. Substitute $n = 2k$ back into the original equation:
$$y[2k] = x[k]$$
6. Rearrange the sides so the input $x$ is isolated on the left:
$$x[k] = y[2k]$$
7. Finally, replace the dummy variable $k$ with the standard discrete-time index $n$:
$$x[n] = y[2n]$$

**Step 4: Final Answer**

The inverse system simply reads the values at the even indices of $y$ and discards the zeros at the odd indices (an operation known as decimation):

$$x[n] = y[2n]$$

[cite: 1]

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